

AMANUL HAQUE

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OVERVIEW

AI Researcher with a strong foundation in generative AI (LLMs), self-supervised representation learning, and multi-agent simulations. Passionate about developing scalable and interpretable AI solutions for real-world problems. Published in leading venues including ACL, IJHCI, IEEE-TCSS, and AI & Society. Visit my [Google Scholar](#) or [Webpage](#) for more details.

EDUCATION

- Doctor of Philosophy (PhD) in Computer Science** • CGPA - **4.0/4** March 2025
North Carolina State University, Raleigh, NC (advisor: Distinguished Prof [Munindar P. Singh](#))
Dissertation: *News, Networks, and Narratives: Language Models and Social Simulations*
- Master of Science in Computer Science** • CGPA - **4.0/4** May 2019
North Carolina State University, Raleigh, NC (advisor: [Dr. Collin F. Lynch](#))
Thesis: *Prioritizing Online Discussion Forum Posts Using Linguistic Features With Limited Labelled Instances*
- Bachelor of Engineering in Information Science** • CGPA - **8.76/10** May 2015
PES University, Bangalore, India

PROFESSIONAL EXPERIENCE

- Data Scientist III - LexisNexis (Nexis + AI), Raleigh** April 2025 - Present
- Designed and implemented a scalable homonym-aware entity deduplication algorithm, expanding coverage by 20x
 - Designed a RAG evaluation framework to assess retrieval quality and answer generation for context engineering
- Summer Research Scholar at SCADs - Laboratory of Analytic Sciences (LAS), Raleigh** June 2024 - July 2024
- Developed a multi-agent LLM-based abstractive summarizer and deployed LLM-as-a-judge for scalable evaluation
 - Demonstrated that multi-agent small models yield more balanced summaries than single large models
- Machine Learning Summer Intern - Coupang, Mountain View** May 2022 - Aug 2022
- Improved Coupang's Deep & Cross Network (DCN) model's efficiency by identifying and removing spurious features
 - Automated benchmarking and built reusable scripts to streamline experiment reproducibility and comparison
- Machine Learning Summer Intern - Seagate, Longmont** May 2020 - Aug 2020
- Designed a graph-based, unsupervised multi-document abstractive summarizer for a social listening app
 - Built an unsupervised aspect-based sentiment analyzer to compare sentiments across online user reviews
- Computer Science Summer Intern - Lenovo, Morrisville** May 2018 - Aug 2018
- Built a log-driven test case prioritization algorithm to identify high failure risk test cases to optimize test execution
 - Reduced manual efforts in testing by automating the extraction of executable commands from test specifications
- Member of Technical Staff - Oracle, Bangalore** July 2015 - June 2017
- Designed and developed Service Deployment Infrastructure (SDI) modules for Oracle Public Cloud (OPC)
 - Designed and implemented reusable and scalable virtual testing environments for SDI services

AWARDS AND HONORS

- Graduate Student Leadership Award 2023 Awarded by NC State University
- Outstanding Teaching Assistant Award 2021 Awarded by NC State University

SKILLS

- Programming Languages: Python, Java, Shell, SQL, PHP, HTML, JavaScript
- API and Libraries: PyTorch, Tensorflow, Langchain, Huggingface, Scikit-learn, Gensim, Numpy, Pandas

LEADERSHIP & COMMUNITY SERVICES

- Reviewed research papers for leading conferences and journals, including ACL, ACM-TIST, AAAI, and AAMAS
- President and Head of Events for the Indian Graduate Student Association (IGSA), NCSU www.maitrincsu.org
- Organizer and Host for the AI in Society Seminar Series at NC State University Talks available on [YouTube](#)
- Mentored several graduate students at NC State University in their research